

Lecture 5: Monte Carlo Methods

Professor Alexander Franks

2025-10-20

Inference for arbitrary functions

- Assume that $Y \sim \text{Bin}(n, \theta)$ but that you are interested in the log odds:

$$\gamma = \log \text{odds}(\theta) = \log \frac{\theta}{1 - \theta}$$

- We use a Beta prior, so that the posterior distribution for θ is also a Beta distribution.

How do we estimate the posterior distribution for the log odds?

Posterior inference for arbitrary functions

Method of transformations

1. Find the inverse, $\theta = g^{-1}(\gamma) = \frac{e^\gamma}{1+e^\gamma}$
2. Compute $\frac{dg^{-1}(\gamma)}{d\gamma}$
3. Find $p_\gamma(\gamma \mid y_1, \dots, y_n) = \left| \frac{dg^{-1}(\gamma)}{d\gamma} \right| \times p_\theta(g^{-1}(\gamma) \mid y_1, \dots, y_n)$

But... don't bother! If we're computing expected values, don't need the method of transformations.

Inference for arbitrary functions

For any $\gamma = g(\theta)$ we have

$$E(g(\theta)|y) = \int g(\theta)p(\theta|y)d\theta$$

Examples:

- $E[\gamma | y] = \int \log(\frac{\theta}{1-\theta})p(\theta | y)d\theta$
- $Pr(\theta \in R | y) = E(I [\theta \in R] | y)$
- $Var(\theta | y) = E [(\theta - E[\theta | y])^2 | y]$

$$\int g(\theta)p_{\theta}(\theta | y_1, \dots, y_n)d\theta = \int \gamma p_{\gamma}(\gamma | y_1, \dots, y_n)d\gamma$$

Law of the Unconscious Statistician



Monte Carlo Integration

Law of the Unconscious Statistician:

$$\int g(\theta) p_{\theta}(\theta \mid y_1, \dots, y_n) d\theta = \int \gamma p_{\gamma}(\gamma \mid y_1, \dots, y_n) d\gamma$$

We can approximate this integral through simulation!

Monte Carlo Integration

- $\overline{\theta} = \sum_{s=1}^S \theta^{(s)} / S \rightarrow E[\theta | y_1, \dots, y_n]$
- $\sum_{s=1}^S \left(\theta^{(s)} - \overline{\theta} \right)^2 / (S - 1) \rightarrow \text{Var}[\theta | y_1, \dots, y_n]$
- $\# \left(\theta^{(s)} \leq c \right) / S \rightarrow \text{Pr}(\theta \leq c | y_1, \dots, y_n)$
- the α -percentile of $\{ \theta^{(1)}, \dots, \theta^{(S)} \} \rightarrow \theta_\alpha$

Monte Carlo Error

- Reminder: $\overline{\theta} = \sum_{s=1}^S \theta^{(s)} / S$ and S is the number of samples.
- If posterior samples are independent then:

$$\text{Var}(\overline{\theta}) = \frac{1}{S^2} \sum_{s=1}^S \text{Var}(\theta^{(s)}) = \frac{\text{Var}(\theta \mid y_1, \dots, y_n)}{S}$$

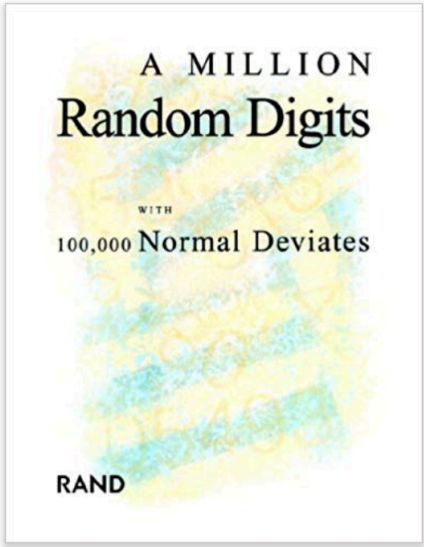
- In general, the Monte Carlo error decreases with $\frac{1}{S}$
- Monte Carlo integration can be very powerful *if* you we can sample from the posterior!
 - This is a *big* “if”

A Million Random Digits with 100,000 Normal Deviates 0th Edition

by [The RAND Corporation](#) (Author)

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1

00000	10097	32533	76520	13586	34673	54876	80959	09117	39292	74945
00001	37542	04805	64894	74296	24805	24037	20636	10402	00822	91665
00002	08422	68953	19645	09303	23209	02560	15953	34764	35080	33606
00003	99019	02529	09376	70715	38311	31165	88676	74397	04436	27659
00004	12807	99970	80157	36147	64032	36653	98951	16877	12171	76833
00005	66065	74717	34072	76850	36697	36170	65813	39885	11199	29170
00006	31060	10805	45571	82406	35303	42614	86799	07439	23403	09732
00007	85269	77602	02051	65692	68665	74818	73053	85247	18623	88579
00008	63573	32135	05325	47048	90553	57548	28468	28709	83491	25624
00009	73796	45753	03529	64778	35808	34282	60935	20344	35273	88435
00010	98520	17767	14905	68607	22109	40558	60970	93433	50500	73998
00011	11805	05431	39808	27732	50725	68248	29405	24201	52775	67851
00012	83452	99634	06288	98083	13746	70078	18475	40610	68711	77817
00013	88685	40200	86507	58401	36766	67951	90364	76493	29609	11062
00014	99594	67348	87517	64969	91826	08928	93785	61368	23478	34113
00015	65481	17674	17468	50950	58047	76974	73039	57186	40218	16544
00016	80124	35635	17727	08015	45318	22374	21115	78253	14385	53763
00017	74350	99817	77402	77214	43236	00210	45521	64237	96286	02655
00018	69916	26803	66252	29148	36936	87203	76621	13990	94400	56418
00019	09893	20505	14225	68514	46427	56788	96297	78822	54382	14598
00020	91499	14523	68479	27686	46162	83554	94750	89923	37089	20048
00021	80336	94598	26940	36858	70297	34135	53140	33340	42050	82341
00022	44104	81949	85157	47954	32979	26575	57600	40881	22222	06413
00023	12550	73742	11100	02040	12860	74697	96644	89439	28707	25815
00024	63606	49329	16505	34484	40219	52563	43651	77082	07207	31790
00025	61196	90446	26457	47774	51924	33729	65394	59593	42582	60527
00026	15474	45266	95270	79953	59367	83848	82396	10118	33211	59466
00027	94557	28573	67897	54387	54622	44431	91190	42592	92927	45973
00028	42481	16213	97344	08721	16868	48767	03071	12059	25701	46670
00029	23523	78317	73208	89837	68935	91416	26252	29663	05522	82562
00030	04493	52494	75246	33824	45862	51025	61962	79335	65337	12472
00031	00549	97654	64051	88159	96119	63896	54692	82391	23287	29529
00032	35963	15307	26898	09354	33351	35462	77974	50024	90103	39333
00033	59808	08391	45427	26842	83609	49700	13021	24892	78565	20106
00034	46058	85236	01390	92286	77281	44077	93910	83647	70617	42941
00035	32179	00597	87379	25241	05567	07007	86743	17157	85394	11838
00036	69234	61406	20117	45204	15956	60000	18743	92423	97118	96338
00037	19565	41430	01758	75379	40419	21585	66674	36806	84962	85207
00038	45155	14938	19476	07246	43667	94543	59047	90033	20826	69541
00039	94864	31994	36168	10851	34888	81553	01540	35456	05014	51176
00040	98086	24826	45240	28404	44999	08896	39094	73407	35441	31880
00041	33185	16232	41941	50949	89435	48581	88695	41994	37548	73043
00042	80951	00406	96382	70774	20151	23387	25016	25298	94624	61171
00043	79752	49140	71961	28296	69861	02591	74852	20539	00387	59579
00044	18633	32537	98145	06571	31010	24674	05455	61427	77938	91936
00045	74029	43902	77557	32270	97790	17119	52527	58021	80814	51748
00046	54178	45611	80993	37143	05335	12969	56127	19255	36040	90324
00047	11664	49883	52079	84827	59381	71539	09973	33440	88461	23356
00048	48324	77928	31249	64710	02295	36870	32307	57546	15020	09994
00049	69074	94138	87637	91976	35584	04401	10518	21615	01848	76938



John Peter O'connor

★★★★☆ **Weirdest sudoku book ever**

October 6, 2012

Format: Paperback

This has got to be the most useless set of sudoku puzzles ever.

★★★★☆ **Not random enough**

By [Patrick Carroll](#) on January 24, 2016

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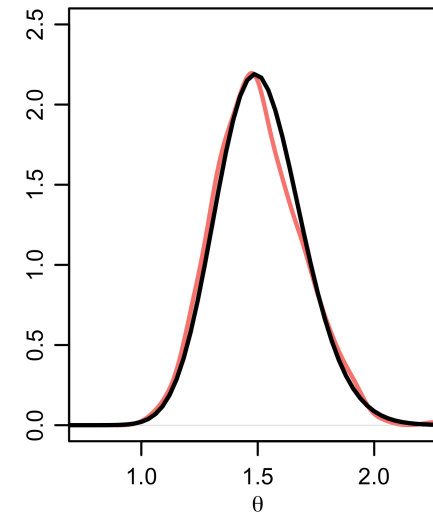
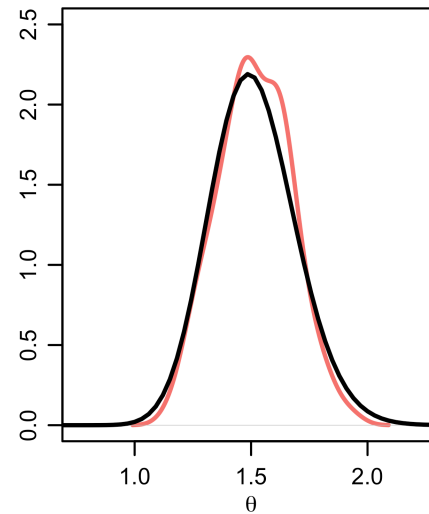
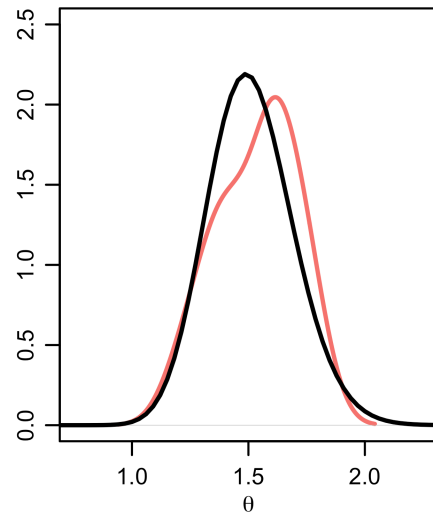
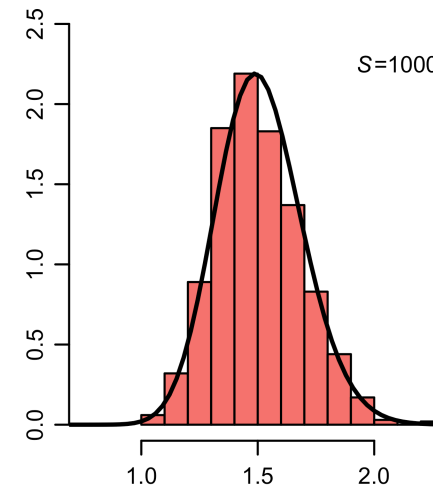
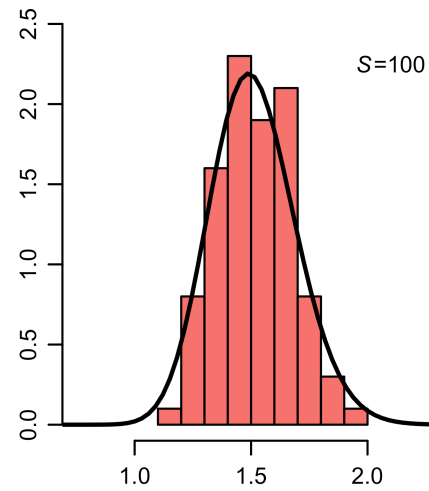
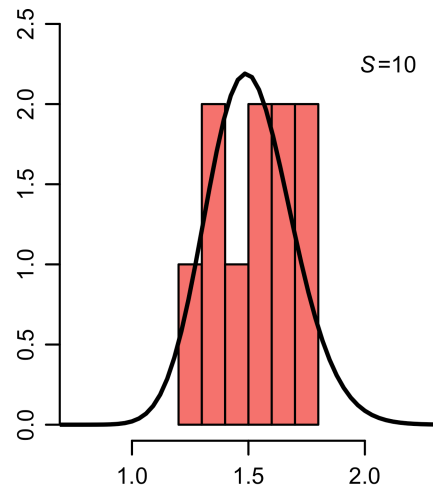
The numbers in the lower left and right corners increase monotonically.

★★★★☆ **Waiting for the movie**

By [Da Vinci](#) on May 23, 2012

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Monte Carlo approximations of a distribution



Inference for arbitrary functions

Assume we want to estimate the posterior mean $E[\gamma \mid y_1, \dots, y_n]$. For example, assume $\gamma = \log \text{odds}(\theta) = \log \frac{\theta}{1-\theta}$. Then:

1. sample $\theta^{(1)} \sim p(\theta \mid y_1, \dots, y_n)$, compute $\gamma^{(1)} = \log\left(\frac{\theta^{(1)}}{1-\theta^{(1)}}\right)$

2. sample $\theta^{(2)} \sim p(\theta \mid y_1, \dots, y_n)$, compute $\gamma^{(2)} = \log\left(\frac{\theta^{(2)}}{1-\theta^{(2)}}\right)$

. sample $\theta^{(3)} \sim p(\theta \mid y_1, \dots, y_n)$, compute $\gamma^{(3)} = \log\left(\frac{\theta^{(3)}}{1-\theta^{(3)}}\right)$

... etc until we have S samples.

Compute $\frac{1}{S} \sum_i^S \gamma^{(i)}$ where S is the number of Monte Carlo samples.

Log-odds of Robert Covington Made 3

Assume $P(\theta \mid y) = \text{Beta}(527, 924)$, the posterior distribution for γ

```
1 theta_samples <- rbeta(1000, 527, 924)
2 gamma_samples <- log(theta_samples / (1-theta_samples))
```

Sampling strategies

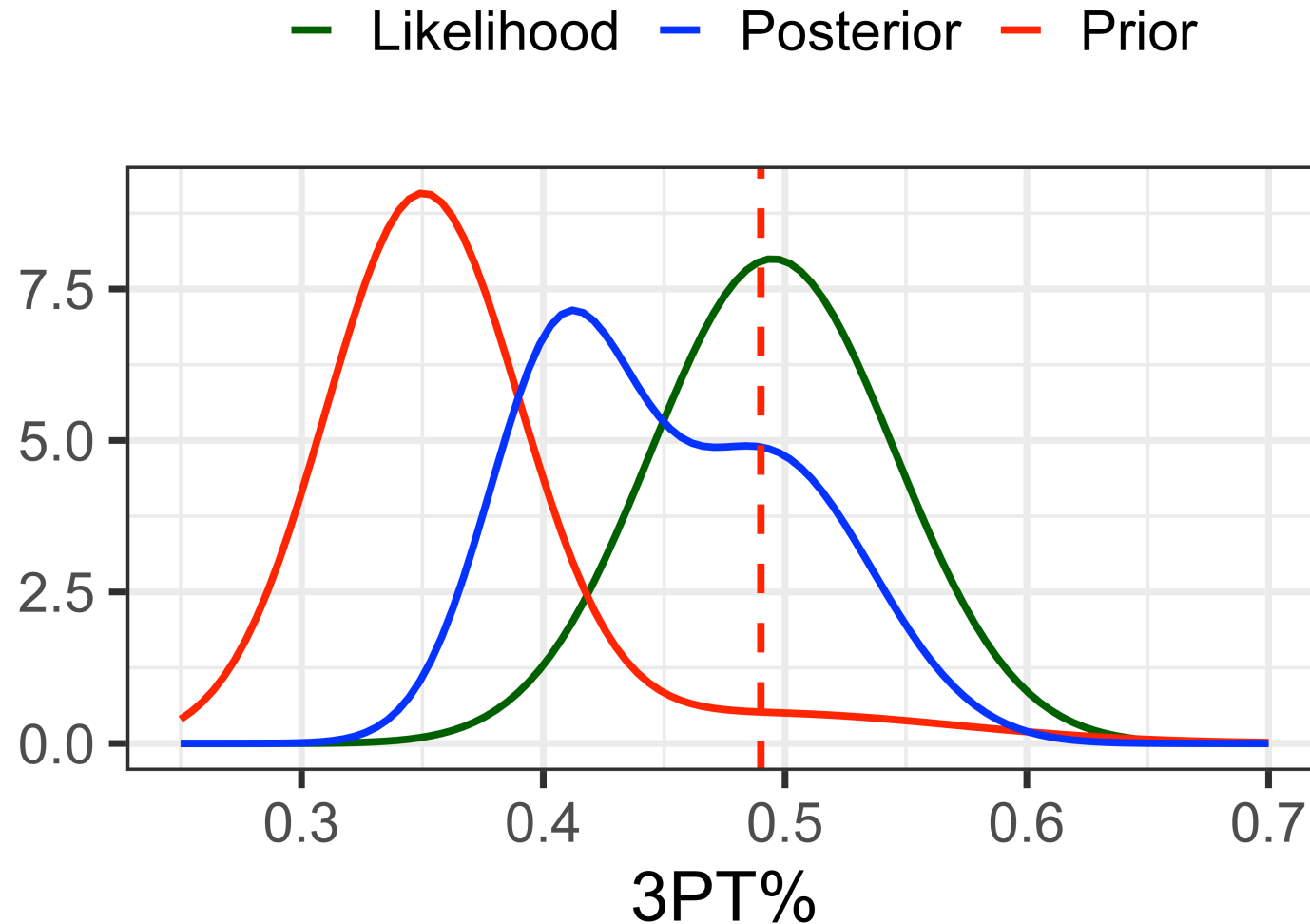
Example: non-conjugate Prior Distributions

- Conjugate prior distributions make the math / concepts easy but no reason they should reflect our true prior belief
- In theory, want to build the best model possible, not one that is convenient
- If we choose a non-conjugate prior distribution, then the posterior distribution may have a “complicated” density. Need Monte Carlo to estimate posterior summaries.

Estimating Robert Covington's skill

- Binomial likelihood is $p(y \mid \theta) \propto \theta^y (1 - \theta)^{n-y}$
- Assume I use a mixture normal prior is
$$p(\theta) = 0.9f_1(\theta) + 0.1f_2(\theta)$$
 - f_1 is $N(\mu = 0.35, \sigma = 0.04)$ and f_2 is $N(\mu = 0.5, \sigma = 0.08)$

Example: estimating three point skill



How can we compute the posterior mean and interval?

Sampling strategies

- Monte Carlo methods assume that we have a method for easily generating a pseudo-random number!
- If the `R` includes the appropriate random number generating function, e.g. `rnorm` then Monte Carlo is easy
- If not, we need to be more clever about how we generate samples.
 - Inversion Sampling (works for univariate)
 - Grid sampling (works for low dimensional problems)
 - Rejection sampling (can be good for low dimensional problems)
 - Importance sampling (useful in some cases, hard in general)
 - Markov Chain Monte Carlo

Sampling strategies

- Reminder: why sampling? We want to approximate difficult integrals.
 - We can represent expected values, probabilities, quantiles etc all as integrals
- In Bayesian stats we usually know how to write down the (proportional) posterior density: $L(\theta)P(\theta)$
- Knowing the pdf does not mean by default we know to sample from that distribution!
- If we can devise a way to sample

Probability Integral Transform

- Suppose that a random variable, Y has a continuous distribution for with CDF is F_Y .
- Then the random variable $U = F_Y(Y)$ has a uniform distribution
 - This is known as the “probability integral transform PIT”
- By taking the inverse of F_Y we have $F^{-1}(U) = Y$

Inversion Sampling

The inverse transform sampling method works as follows:

1. Generate a random number u from $\text{Unif}[0, 1]$
2. Find the inverse of the desired CDF, e.g. $F_Y^{-1}(u)$.
3. Compute $y = F_Y^{-1}(u)$. y is now a sample from the desired distribution.

Inversion Sampling

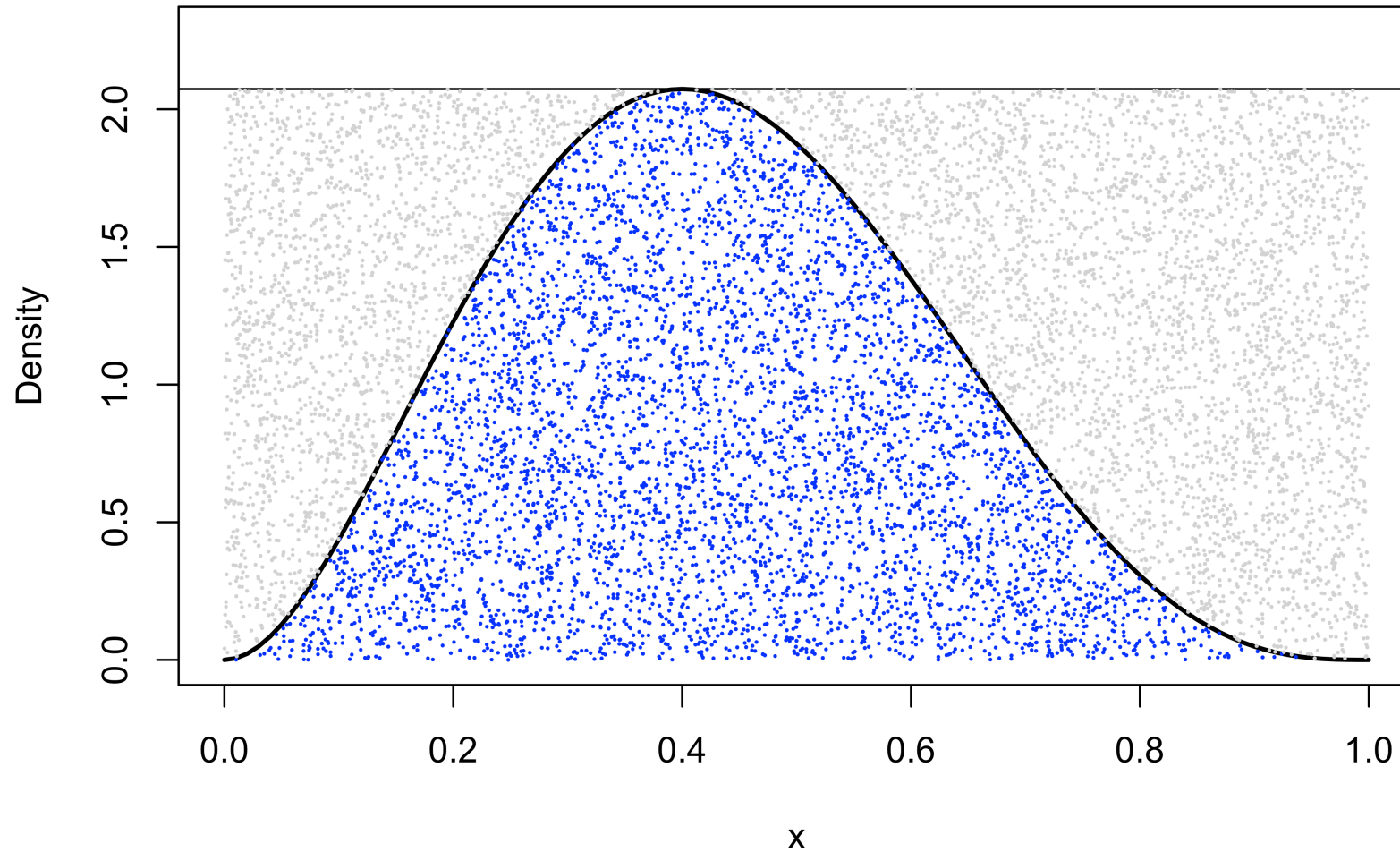
Animation Demo

Inversion Sampling

- Inversion sampling can be a fast and simple way to sample from a distribution
- Only effective if we know the inverse-CDF and can easily compute it
- This is a big challenge in practice. For example, even the normal distribution has a CDF, Φ , which cannot be expressed analytically.
 - Shifts from one hard problem (sampling) to another (computing an integral)
 - Need alternatives!

Rejection Sampling

Acceptance rate: 0.486600



Rejection Sampling algorithm

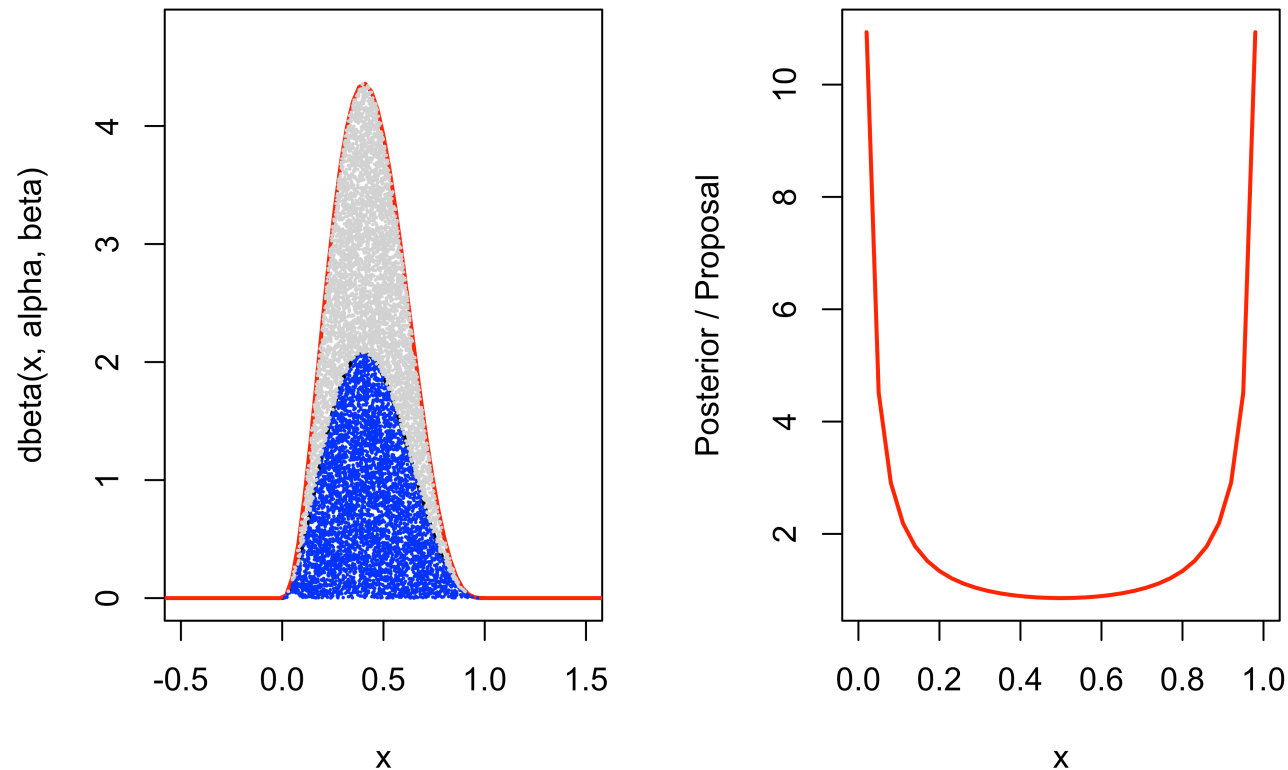
1. Choose a proposal density, $q(\theta)$ that we can easily sample from (e.g. uniform or normal) such that:
2. Find $M = \max \frac{p(\theta|y)}{q(\theta)}$
 - If $M = \infty$ then q cannot be used as a proposal distribution
 - If M is finite, $Mq(\theta)$ “envelopes” $p(\theta|y)$
3. Draw a sample, $\theta^{(s)}$ from $q(\theta)$
4. Accept $\theta^{(s)}$ as a draw from $p(\theta|y)$ with probability $\frac{p(\theta^{(s)}|y)}{Mq(\theta^{(s)})}$

Rejection Sampling

Demo

Proposal most envelope target

Sometimes its not obvious...



Target density in black, proposal density in red

Markov Chain Monte Carlo

- Markov Chain Monte Carlo (MCMC)
- More effective approach to sampling from multi-parameter distributions
- Samples in MCMC are **not** independent samples

MCMC Sampling with Stan

Stan



“A state-of-the-art platform for statistical modeling and high-performance statistical computation.”

<http://mc-stan.org/>

Monte Carlo Sampling with Stan

- Stan is a “probabilistic programming language” with its own simple syntax
- R interface to Stan via `cmdstanr` package
- Define parameters θ and datatypes y
- Provide sampling model $p(y \mid \theta)$ and prior $p(\theta)$
- Stan translates model syntax into the log posterior density $\log(p(\theta \mid y))$ and automatically generates samples from this density

MCMC with Stan

```
1 data {  
2   int<lower=0> makes;  
3   int<lower=makes> attempts;  
4 } parameters {  
5   real<lower=0, upper=1> theta;  
6 }  
7 model {  
8   makes ~ binomial(attempts, theta);  
9   theta ~ uniform(0, 1);  
10 }
```

MCMC with Stan

Beta Prior

```
1 data {  
2   int<lower=0> makes;  
3   int<lower=makes> attempts;  
4 } parameters {  
5   real<lower=0, upper=1> theta;  
6 }  
7 model {  
8   makes ~ binomial(attempts, theta);  
9   theta ~ beta(30, 70);  
10 }
```


MCMC with Stan

```
# A tibble: 2 × 10
  variable      mean  median      sd    mad      q5      q95  rhat  ess_bulk
  <chr>      <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl> <dbl>   <dbl>
1 lp__      -136.    -135.    0.706  0.302  -137.   -135.    1.00   1999.
2 theta       0.398    0.397  0.0338 0.0341    0.343    0.454    1.00   1432.
# i 1 more variable: ess_tail <dbl>
[1] 0.39801
```

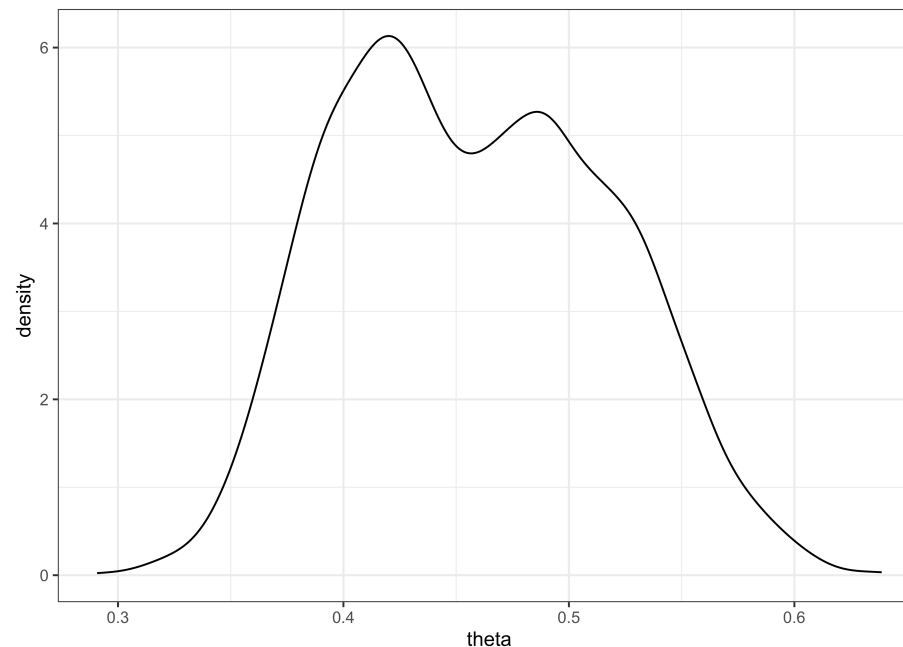
Mixture Prior

```
1 data {  
2   int<lower=0> makes;  
3   int<lower=makes> attempts;  
4 } parameters {  
5   real<lower=0, upper=1> theta;  
6 }  
7 model {  
8   makes ~ binomial(attempts, theta);  
9   // theta ~ 0.9 * N(.35, sigma=0.04) + 0.1 * N(.5, sigma=0.08)  
10  target += log_mix(0.9, normal_lpdf(theta | 0.35, 0.04), normal_lpdf(theta  
11 }
```

Mixture Prior

```
1 # BEGIN SOLUTION
2 stan_fit <- binmod_mix_prior$sample(data=list(makes = 50, attempts=101),
3                                     refresh=0, show_messages = FALSE)
4 samples <- stan_fit$draws(format="df")
5 # END SOLUTION
6
7 samples %>% ggplot() + geom_density(aes(x=theta)) + theme_bw()

1 post_mean <- mean(samples$theta)
```



Posterior Predictive Checks

Posterior predictive model checking

- Let y_{obs} represent the observe data $y_1, \dots, y_n$
- Let $y\sim$ represent n replicated (e.g fake) observations generated from the model
- $p(y\sim | y_{\text{obs}}) = \int p(y\sim | \theta) p(\theta | y_{\text{obs}}) d\theta$
- Generate test quantity from $t(y\sim)$
- Check if the simultaed test quantities are similar to the observed test quantity, $t(y_{\text{obs}})$

Posterior predictive model checking

- If the model fits the data, then fake data generated under the model should look similar to the observed data
- Discrepancies can be due to model misfit or chance (or both!)
- Monte Carlo approach:
 1. sample $\theta^{(s)} \sim p(\theta|Y = \mathbf{y}_{\text{obs}})$
 2. sample $\mathbf{y}^{(s)} = (y_1^{(s)}, \dots, y_n^{(s)}) \sim \text{i.i.d. } p(y|\theta^{(s)})$
 - $\mathbf{y}$ has same number of observations as $\mathbf{y}_{\text{obs}}$
 3. compute $t^{(s)} = t(\mathbf{y}^{(s)})$

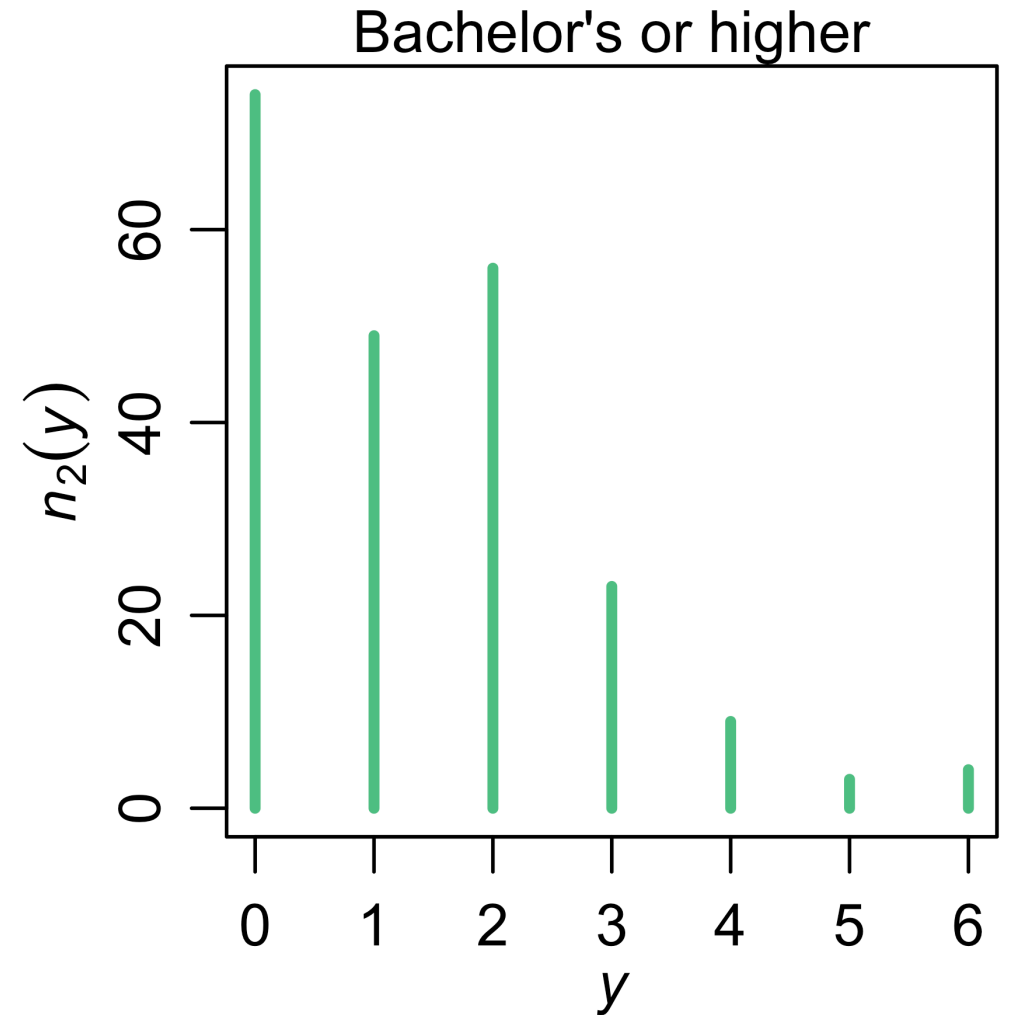
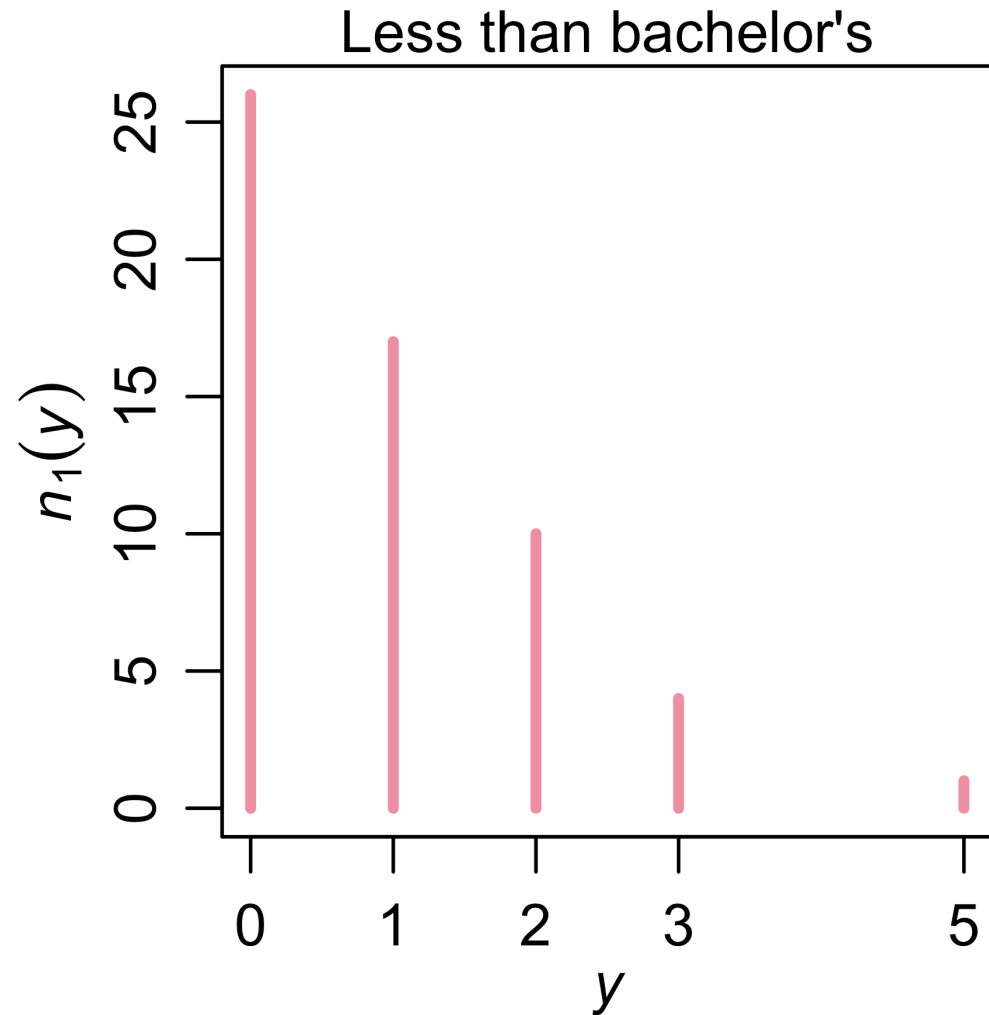
Predictive Checks: an example

- In the 1990's there was a survey of 155 women, at least 40 years of age
- Recorded number of children and educational attainment
 - Bachelor's degree or higher ($n_1 = 111$)
 - Less than bachelor's degree ($n_2 = 44$)

$$Y_{1,1} \dots, Y_{n_1,1} | \theta_1 \sim \text{i.i.d. Poisson}(\theta_1)$$

$$Y_{1,2} \dots, Y_{n_2,2} | \theta_2 \sim \text{i.i.d. Poisson}(\theta_2)$$

PPCs example



A Bayesian Modeling Process (overview)

1. Propose a sampling model or DGP, here $Y \sim \text{Pois}(\theta)$
2. Propose a prior distribution, here $\theta \sim \text{Gamma}(a, b)$
3. Compute the posterior distribution, here $p(\theta \mid Y = y) \sim \text{Gamma}(a + y, \beta + \nu)$
4. Simulate test statistics, $T(y^{(s)})$ from the posterior predictive distribution
 - for s in $1:\text{num_fake_data}$
 - Sample $\theta^{(s)} \sim \text{Gamma}(a + y, b + \nu)$
 - Sample $y^{(s)} \sim \text{i.i.d Poiss}(\theta^{(s)})$ (same sample size as y_{obs})
 - Compute $T(y^{(s)})$
5. Compare the samples $T(y^{(s)})$ to $T(y_{\text{obs}})$
6. Identify any model misfit, go back to step 1 and repeat.

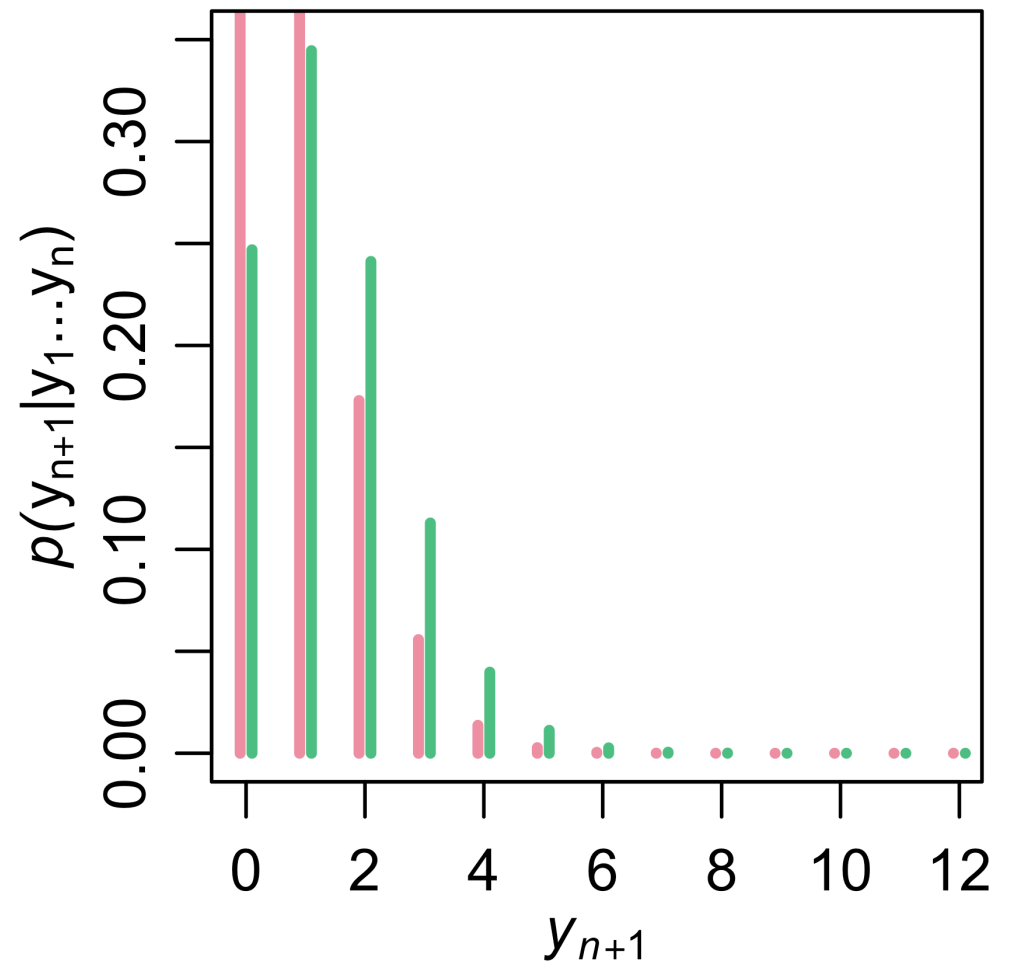
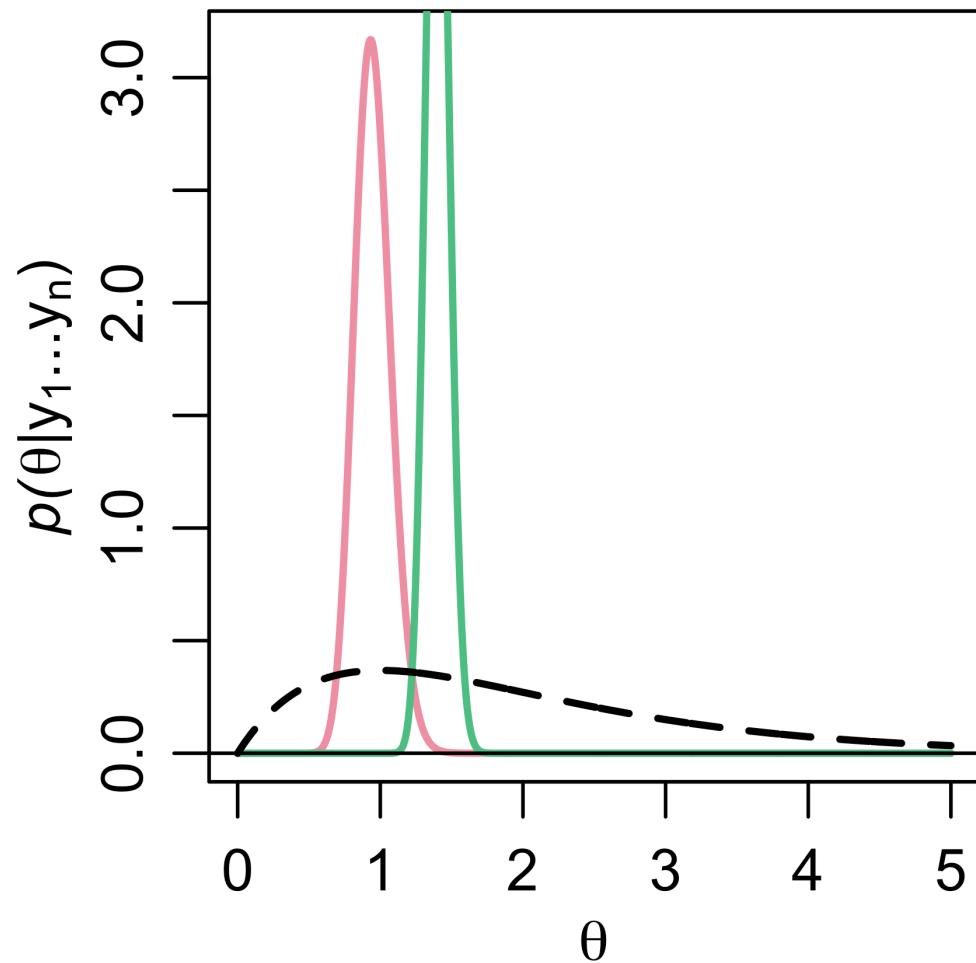
PPCs example

- In the 1990's there was a survey of 155 women, at least 40 years of age
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 - Bachelor's degree or higher ($n_1 = 111$)
 - Less than bachelor's degree ($n_2 = 44$)

$$Y_{1,1} \dots, Y_{n_1,1} | \theta_1 \sim \text{i.d. . Poisson } (\theta_1)$$

$$Y_{1,2} \dots, Y_{n_2,2} | \theta_2 \sim \text{i.d. Poisson}(\theta_2)$$

PPCs example



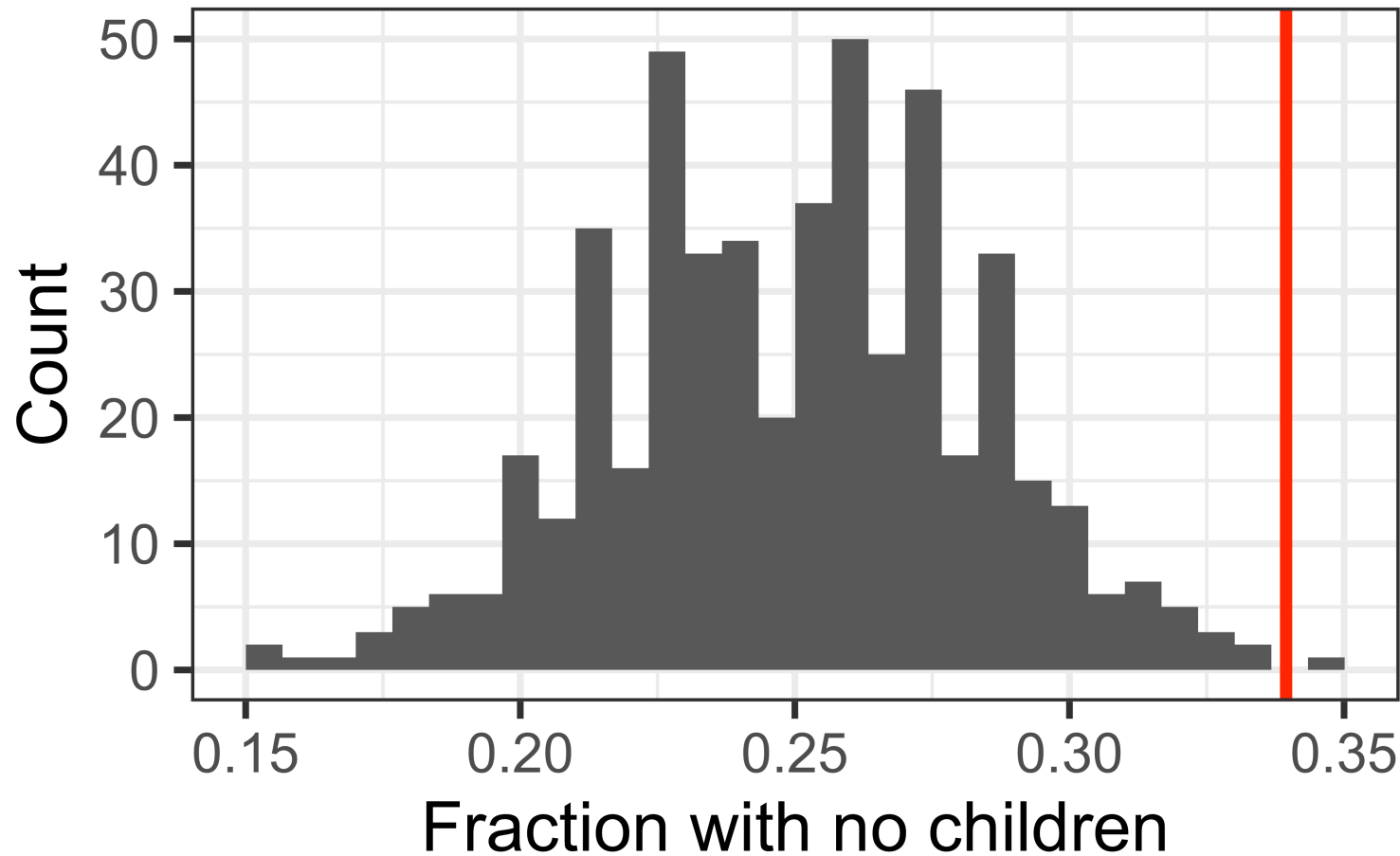
PPCs example

- Let's check the model fit for the “without Bachelor's” group first
- Do S times:
 - sample $n_2 = 44$ observations $\tilde{y}$ from the posterior predictive distribution
- Let $T(\tilde{y})$ be the fraction of women with no children

```
1 S <- 1000
2 t_s <- numeric(S)
3 for(s in 1:S){
4   theta_s <- rgamma(1, a, b) # whatever a and b are for my posterior
5   ytilde_s <- rpois(n=44, theta = theta_s)
6   t_s[s] <- mean(ytilde_s == 0) # compute test stat
7 }
8
9 ## then visualize histogram of t_s
```

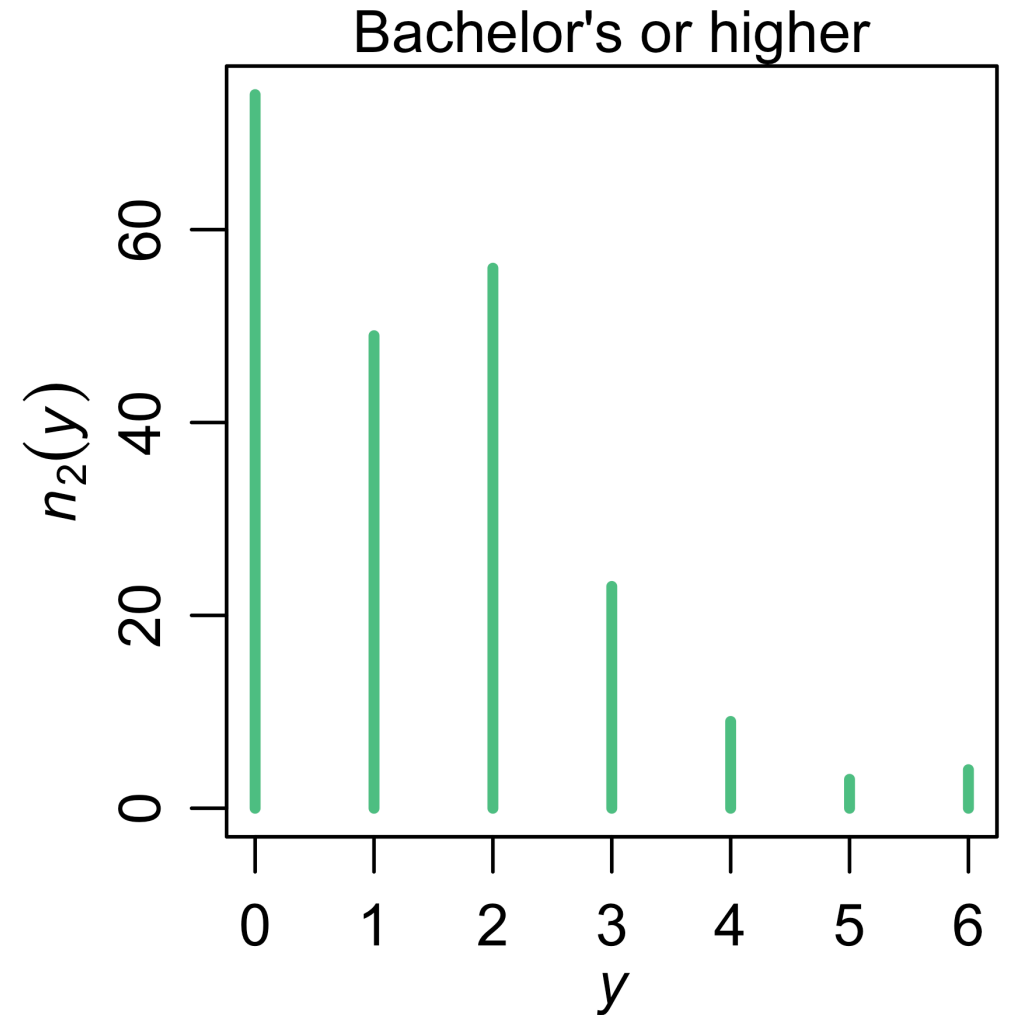
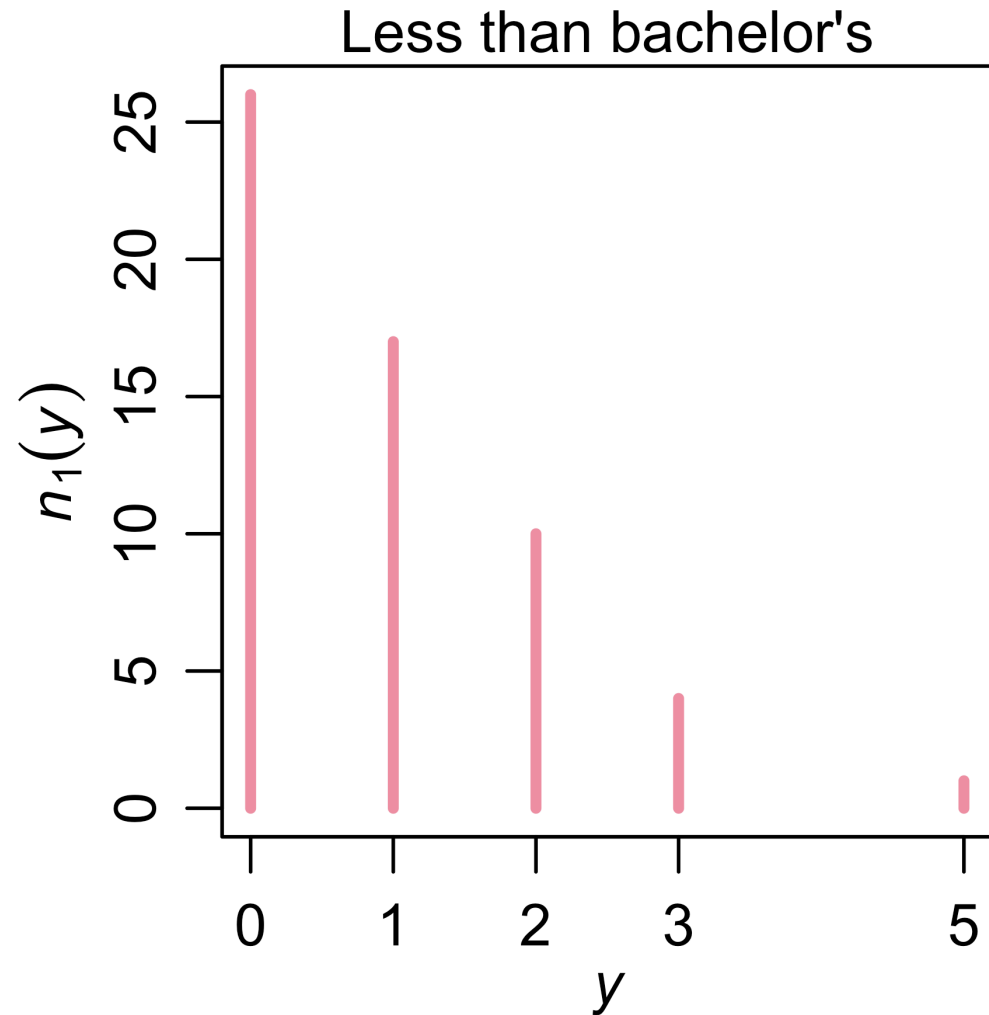
PPCs example

Women without Bachelors degree



$$Pr(T^{\text{rep}} > T^{\text{obs}}) = 0.002$$

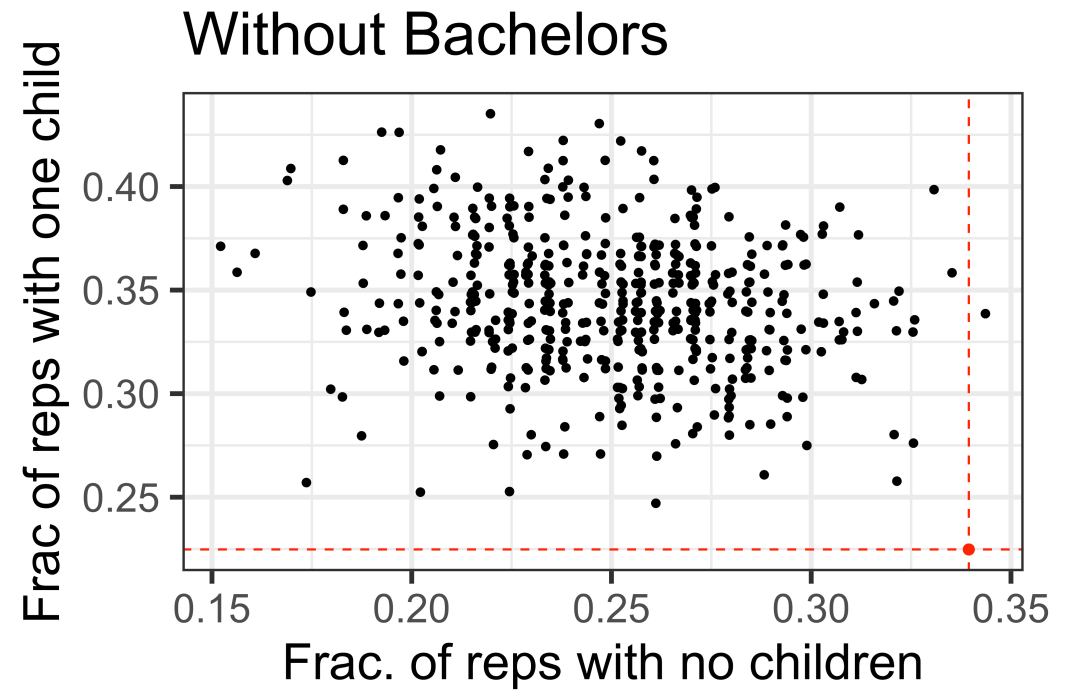
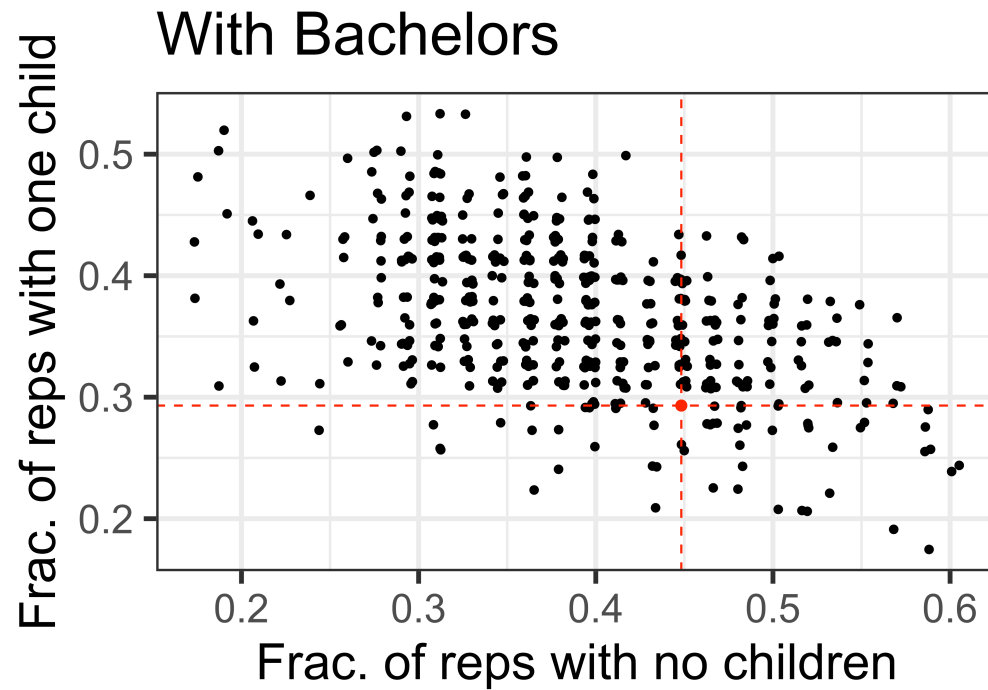
PPCs example



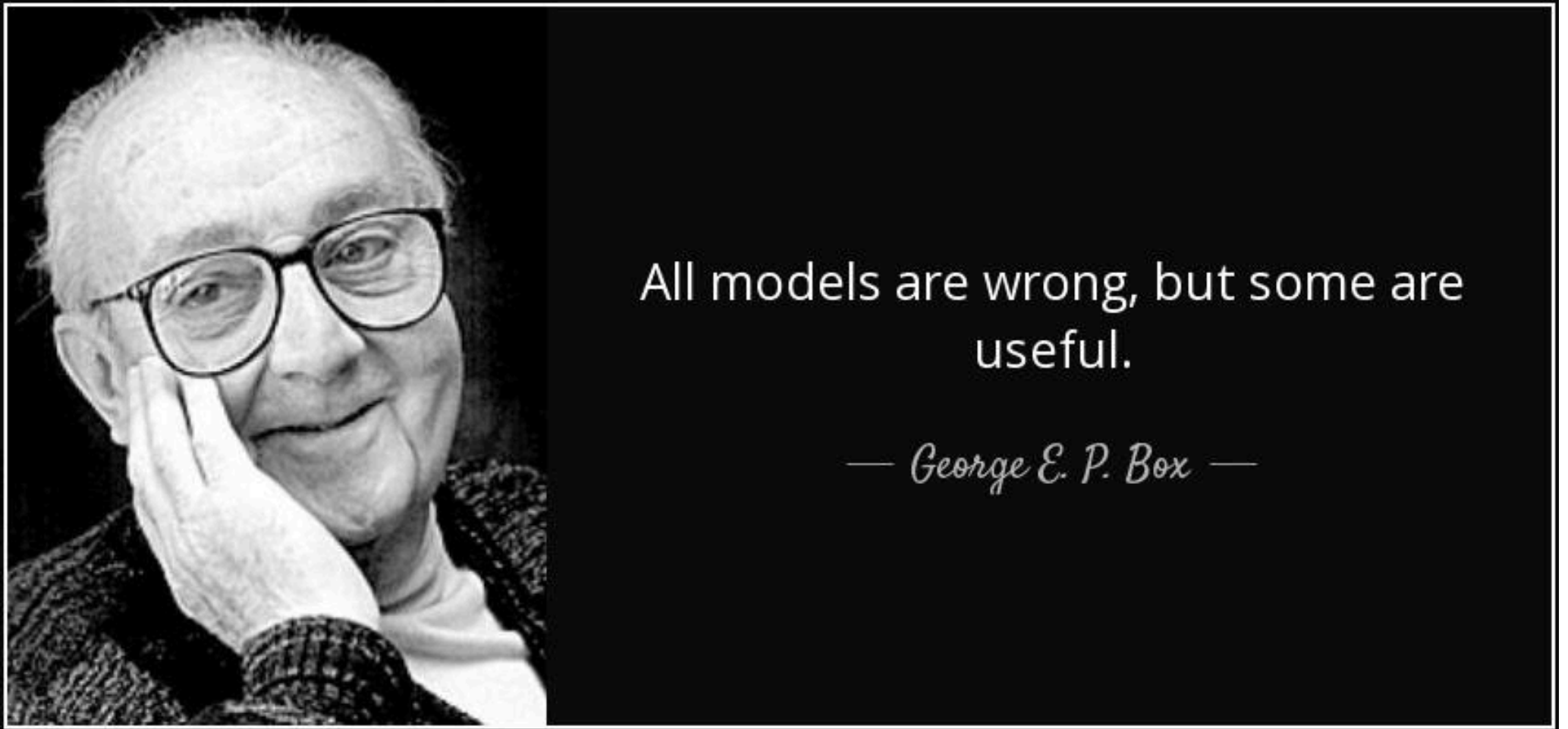
PPCs example

- Model checking both groups
- Look at fit for two different test statistics:
 - Fraction with no children
 - Fraction with one child

Poisson example



All models are wrong



If the model is “wrong”, how can we improve it?

PPCs and Model Refinement

- How might we refine the model?
- What might be a better data generating process?
- How do we choose test statistics to investigate? What other statistics might be worth checking?